Why the GRC MCSI V4 System is Better Than Peter's Method

A data-driven comparison using 17,024 Kings & Queens range scores (2024–2026, 7 state associations) and four seasons of GRC club data

Executive Summary

The GRC MCSI V4 system produces more equitable cross-discipline scores than Peter's method. Peter's formula systematically over-rewards F-class shooters relative to Target Rifle and Sporter shooters: across 121 big Kings & Queens matches, F-class disciplines win **97%** of matches under Peter's, while TR and Sporter combined win **2%**. V4 reduces that imbalance considerably (F-class 65%, TR + Sporter 35%) and produces median scores that are 2.6× tighter across disciplines.

This document presents five arguments, each backed by actual K&Q data — not normalised projections.

Argument 1: Peter's Formula Heavily Favours F-Class at K&Q Level

Across 17,024 individual range scores from K&Q events 2024–2026:

Discipline	n	Median raw	Median Peter MCSI	Median V4 MCSI
F-Open	3,465	59.10	86.18	86.90
F-Standard	3,278	57.20	85.80	85.85
FTR	2,069	57.40	83.40	85.69
Sporter Production	963	49.50	82.62	87.56
Sporter Open	720	49.70	80.85	88.20
TR	6,529	48.70	79.56	86.48

Peter's median spread: 6.62 points (F-Open 86.18 → TR 79.56)
V4 median spread: 2.50 points (Sporter Open 88.20 → FTR 85.69)

V4's spread is **2.6× tighter** than Peter's. Under Peter's, the median TR shooter scores **6.6 points below** the median F-Open shooter for an equivalent placing within their discipline.

Argument 2: Peter's Hands K&Q Wins Almost Exclusively to F-Class

Of the 121 big K&Q matches (≥20 shooters) in 2024–2026:

Discipline	V4 wins	Peter's wins
F-Open	8%	40%
F-Standard	45%	54%
FTR	12%	3%
TR	26%	2%

Discipline	V4 wins	Peter's wins
Sporter Open	3%	0%
Sporter Production	6%	0%
F-class combined	65%	97%
TR + Sporter combined	35%	2%

Under Peter's, a TR or Sporter shooter has effectively no path to a K&Q match win regardless of how well they shoot. V4 still leans F-class but leaves meaningful room for the other disciplines.

Argument 3: Peter's Method Conflates Scale Conversion with Equipment Difficulty

V4 separates two distinct corrections:

1. Score Conversion (1.2): TR, SO, SP shoot a 50-point string; F-class shoot a 60-point string. Multiplying TR/Sporter raw scores by 1.2 puts everyone on the same 60-point scale before any equipment correction.

2. Centre Bonus (0.7): Partial credit for V/X precision without letting centres dominate.

3. Equipment Factor: Per-discipline correction for equipment advantage.

Peter's formula *(raw + centres)* × *factor* collapses scale conversion, centre weighting and equipment factor into a single multiplier. The same factor is doing all three jobs at once, so it cannot be tuned for one without changing the others.

A practical consequence: under Peter's, a 50-point TR perfect produces *(50 + 10) × 1.53 = 91.80*. A 60-point F-Open perfect produces *(60 + 10) × 1.39 = 97.30*. The F-Open shooter ends 5.5 points ahead **at the very ceiling** — not because F-Open is harder, but because the 50→60 scale gap is baked into the factor.

Argument 4: Peter's Weights Centres as Full Points; V4 Weights Them as 0.7

Peter's effectively gives every centre a full point of score (since it's added to raw before multiplying). V4 weights centres at 0.7, recognising precision without letting it dominate the numeric score.

Why this matters: F-class shooters at elite K&Q level routinely hit 60.7–60.8 (7–8 X's). TR shooters routinely hit 50.9–50.10. The full-centre weighting in Peter's combined with F-class's higher raw base compounds the F-class advantage.

Argument 5: V4 is Transparent and Reproducible

V4 has three explicit, auditable parameters:

Parameter	Value	Purpose
Score Conversion	1.2 (TR/Sporter)	Normalises 50→60 scale
Centre Bonus	0.7 per centre	Partial precision credit
Equipment Factor	1.40 – 1.46	Equipment difficulty adjustment

Each step is independently adjustable. Any member can verify their score in four lines of arithmetic.

Peter's uses a single combined multiplier per discipline with no decomposition. It cannot be adjusted for one component without affecting the others, and there is no documented derivation showing where the factors came from.

A Note on Honest Limitations

V4 is not perfect. On the same 121-match K&Q dataset:

• F-Standard still wins 45% of matches under V4 (over-represented vs an ideal ~17%)
• Sporter Open wins only 3%

V4 is **better than Peter's**, but it is not yet fully balanced. The remaining bias is structural — F-class shooters earn more centres per perfect shoot than TR/Sporter shooters can, and the 0.7 centre bonus rewards that disproportionately. Future iterations may need to drop the centre weight further (0.7 → 0.4–0.5) or apply per-discipline centre weights.

Summary

Criterion	V4	Peter's
Median MCSI spread across disciplines (K&Q data)	2.50 pts	6.62 pts
F-class share of big K&Q match wins	65%	97%
TR + Sporter share of big K&Q match wins	35%	2%
Score-conversion methodology	Separate, explicit	Baked into multiplier
Centre bonus weight	0.7 (partial)	1.0 (full point)
Reproducibility	Three auditable parameters	Single multiplier
Honest about residual bias	Yes (F-Std still 45%)	Not addressed

V4 is the better current option. Both formulas still over-favour F-class at the elite K&Q level — V4 just does so much less than Peter's.

Analysis based on Kings & Queens 2024–2026 data: 17,024 individual 10-shot range scores from NSWRA, VRA, QRA, NQRA, SARA, WARA and NRAA championships, plus 270 GRC club results across four shoots in 2026.